

Adiponectin expression in subcutaneous adipose tissue is reduced in women with cellulite.



BACKGROUND: Cellulite, which appears as orange peel-type or cottage cheese-like dimpling of the skin on the thighs and buttocks, is a complex, multifactorial, cosmetic disorder of the subcutaneous fat layer and the overlying superficial skin. Adiponectin is an adipocyte-derived hormone mainly produced by subcutaneous fat that shows important protective anti-inflammatory and vasodilatory effects. We hypothesized that adiponectin expressed in the subcutaneous adipose tissue (SAT) might play a role in the pathogenesis of cellulite. We reasoned that a reduction in the expression of adiponectin - a humoral vasodilator - in the SAT of cellulite areas might contribute to the altered microcirculation frequently found in these regions. **METHODS:** A total of 15 lean (body mass index [BMI] < 25 kg/m²) women with cellulite and 15 age- and BMI-matched women without cellulite participated in this study. Real-time reverse transcription polymerase chain reaction (RT-PCR) was used to assess adiponectin gene expression. Plasma adiponectin levels were measured using a commercial enzyme immunoassay kit. **RESULTS:** Adiponectin mRNA expression in the SAT of the gluteal region was significantly lower in areas with cellulite compared with those without (12.6 ± 3.1 AU versus 16.6 ± 4.1 AU; $P=0.006$). However, plasma adiponectin levels did not differ between women with (20.3 ± 7.3 µg/ml) and without (19.3 ± 6.1 µg/ml) cellulite ($P=0.69$). **CONCLUSIONS:** Adiponectin expression is significantly reduced in the SAT in areas affected by cellulite. Our findings provide novel insights into the nature of cellulite and may give clues to the treatment of this cosmetic issue. © 2011 The International Society of Dermatology.